

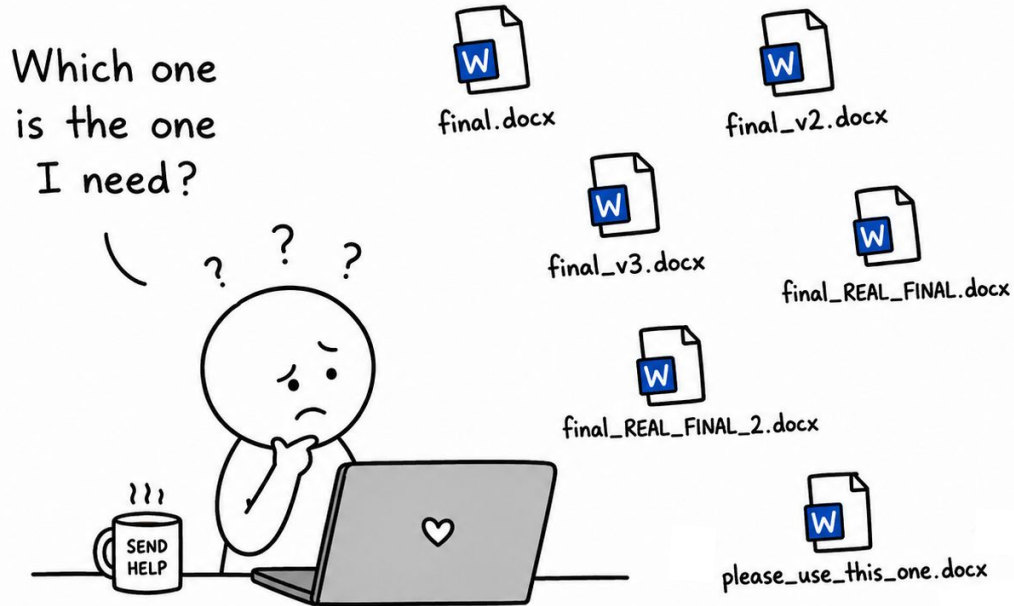


Play with

Git

Wanyi (“One E”) Tang
wanyitan@usc.edu
USC LA’s BeST Program
06/16/26

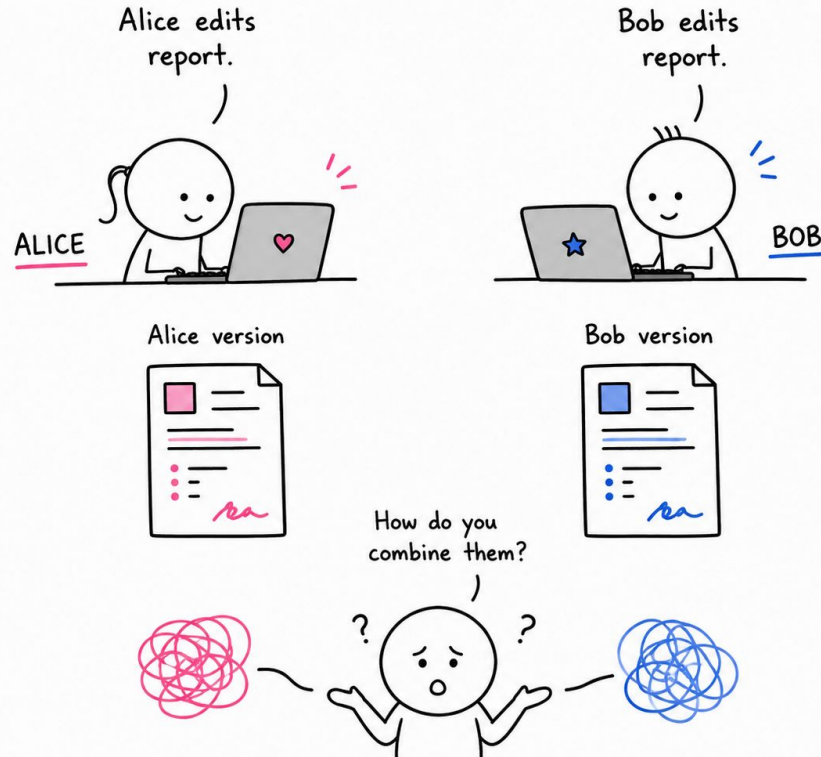
Scenario 1: Which version?



Scenario 2: How to go back?



Scenario 3: How to collab

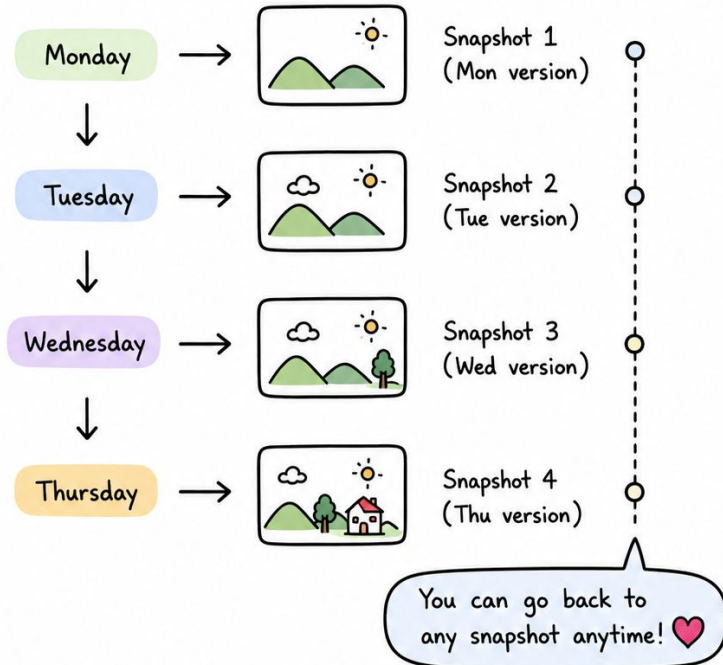
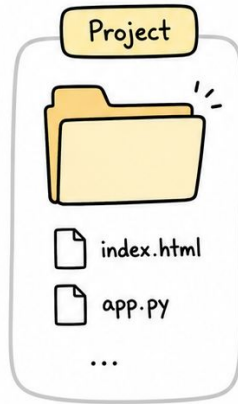
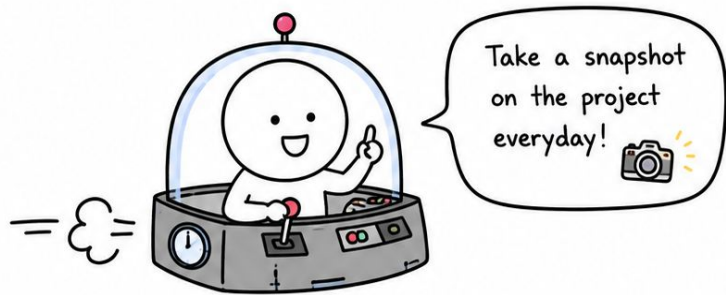


Git can help with all these scenarios!



**git can remember
every change
you make in a project**

Think Git as a time machine





vs.



Lives on your computer

Lives on the internet

Tracks changes

Stores and shares projects

GitHub is like the Google Drive for Git projects.

5 things  **git** can do

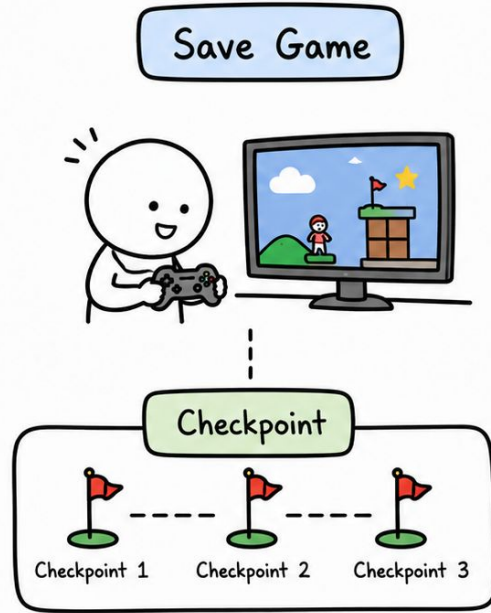
Save progress?



Save

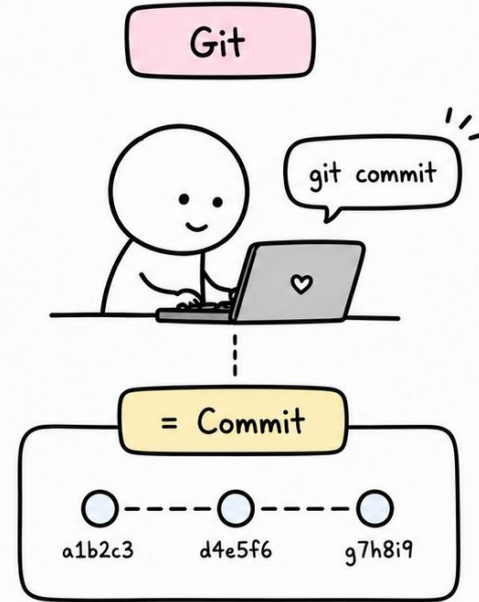
Cancel

Thing 1: Save checkpoint



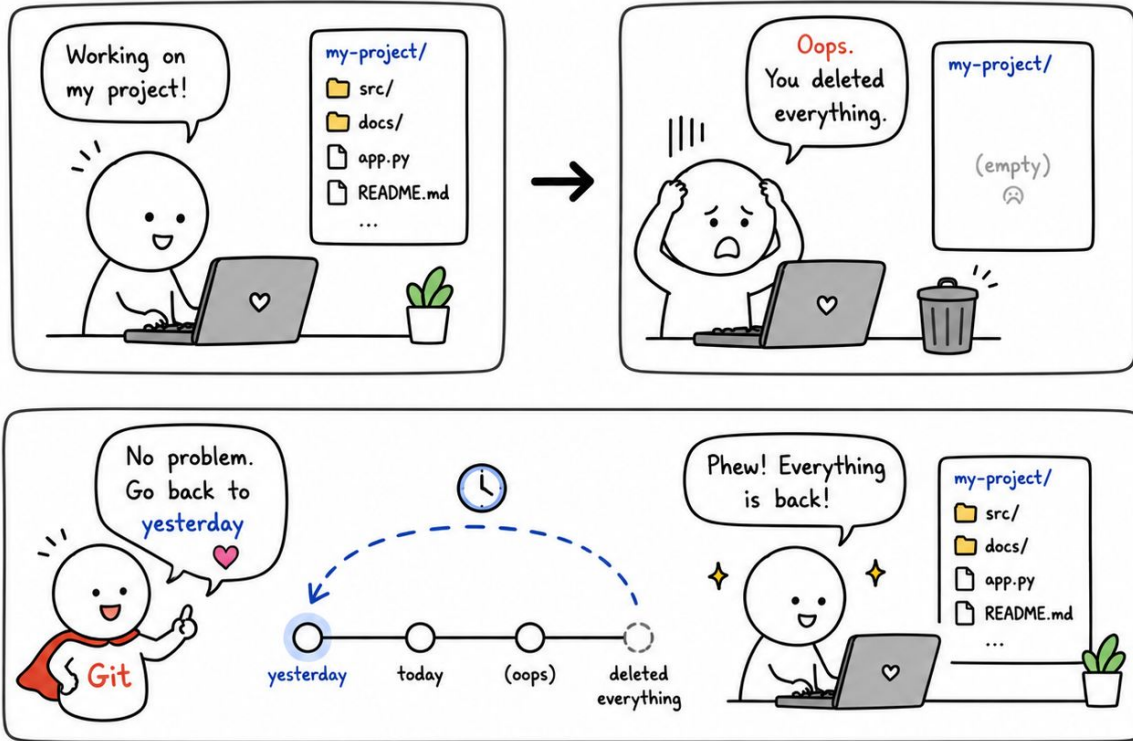
You can go back to any
checkpoint anytime! ❤️

=

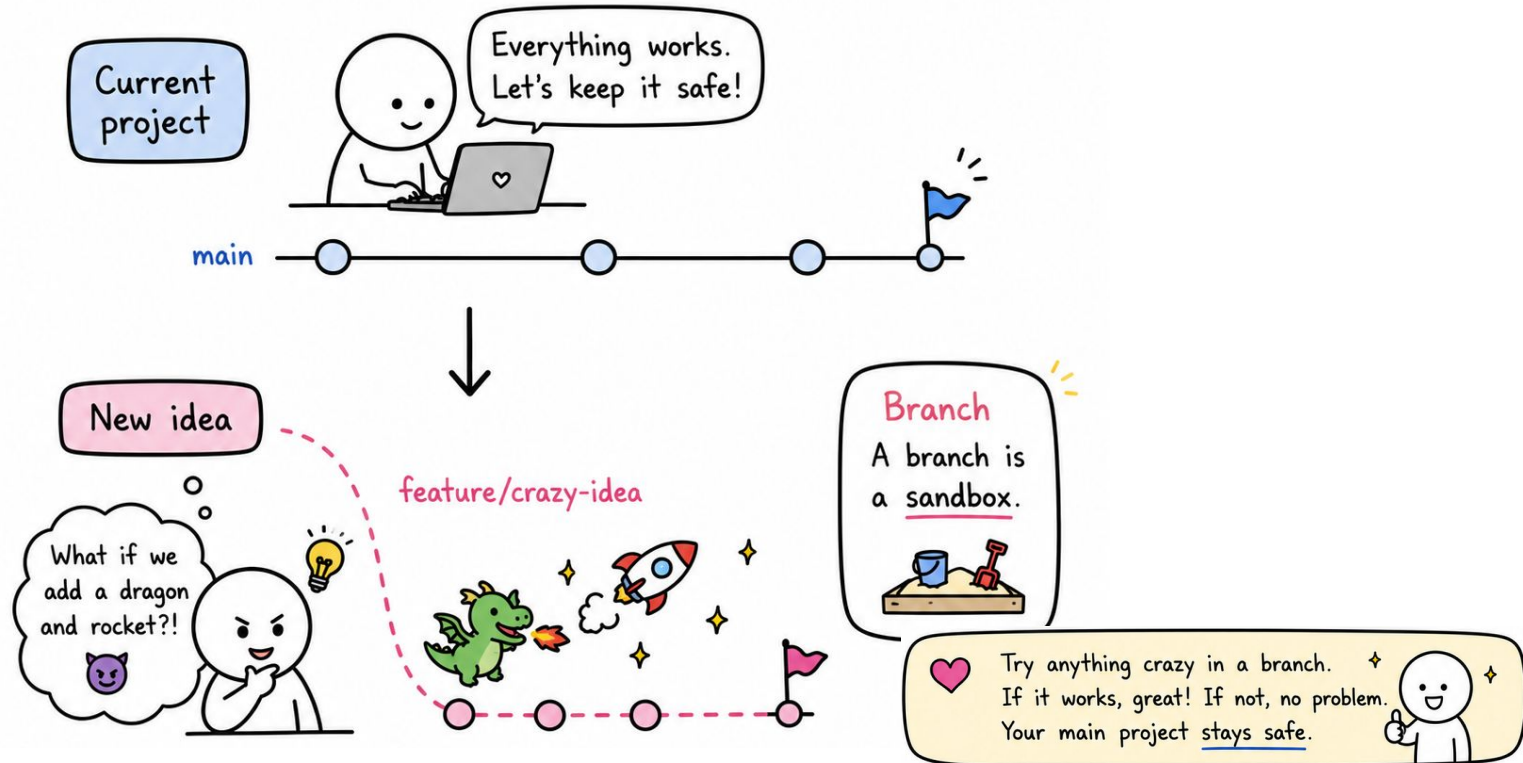


You can go back to any
commit anytime! ❤️

Thing 2: Time travel

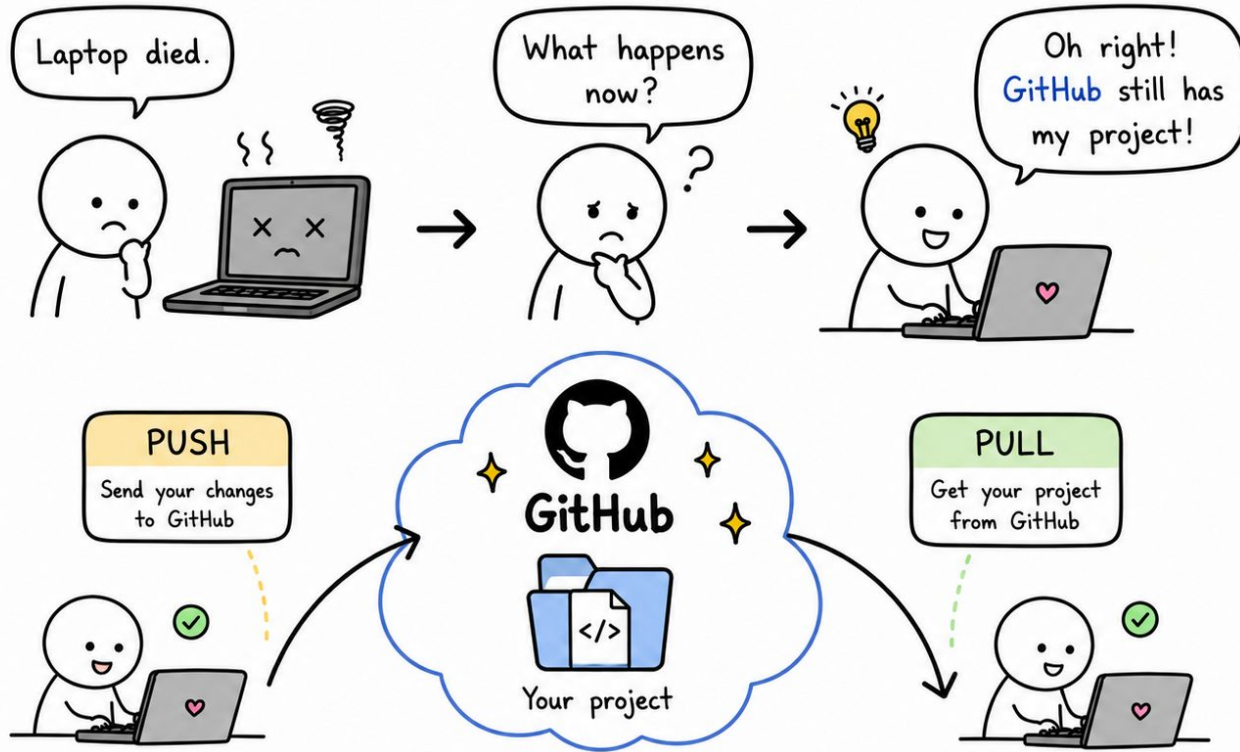


Thing 3: Experiment safe

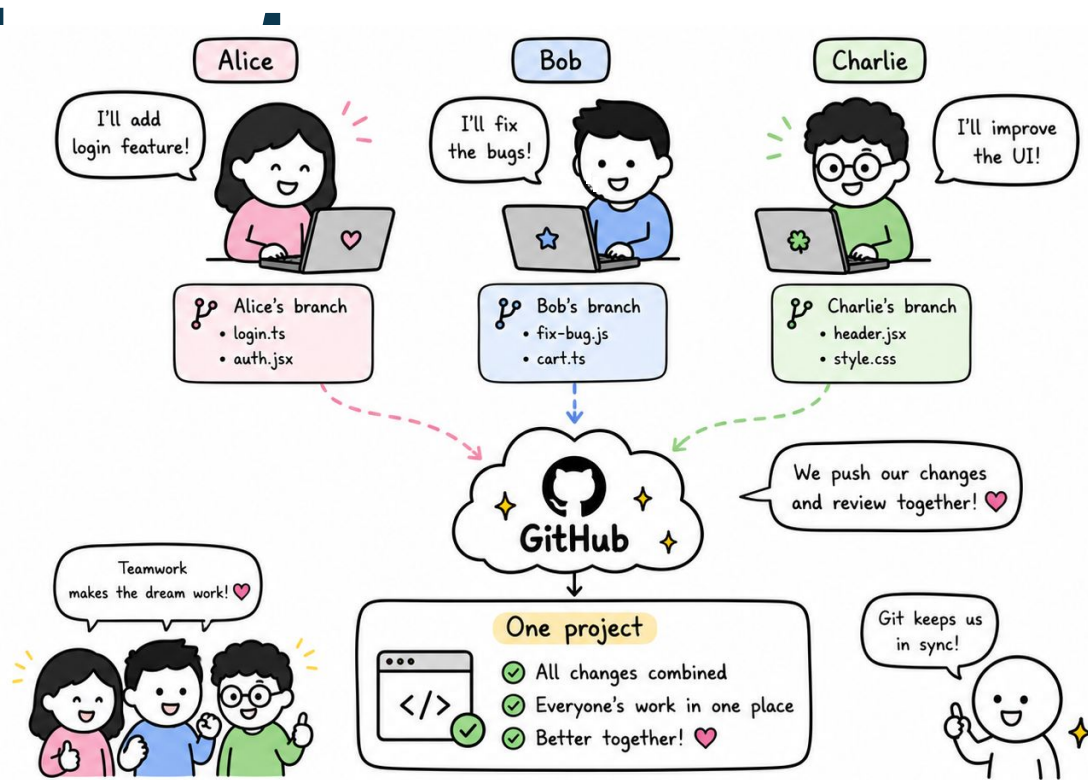


Thing 4: Backup your

prc



Thing 5: Help you colla'



Before we start:

GEEK TERM	HUMAN TRANSLATION
Repository	Project Folder 
Commit	Save Checkpoint 
Branch	Safe Sandbox 
Clone	Download Project 
Push / Pull	Upload / Download from GitHub 



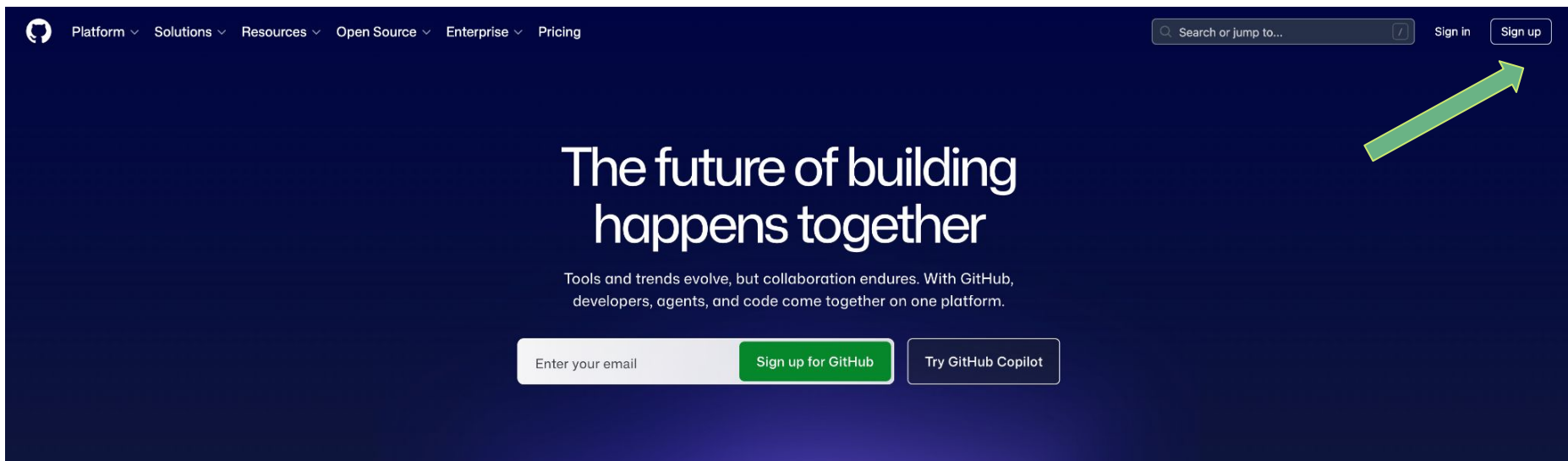
is not an app

is a version control tool

You can use it on lots of platforms

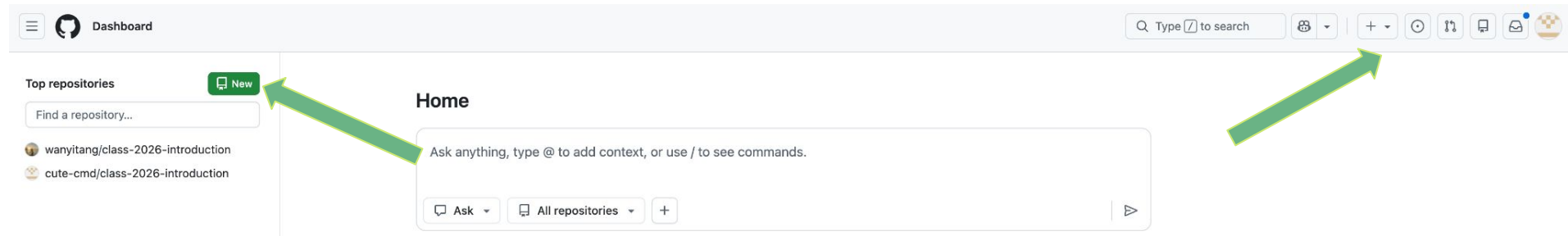
Step 1: Create a GitHub account

Google: github



Create a repository

Create a repository in your github.com



Create a repository

Create a repository in your
github.com

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).

Required fields are marked with an asterisk (*).

1 General

Owner *

 wanyitang ▾

Repository name *

TEST

✓ TEST is available.

Great repository names are short and memorable. How about [laughing-lamp](#)?

Description

This a test repository for LA's BeST program

44 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

 Public ▾

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

On 

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore ▾

Add license

Licenses explain how others can use your code. [About licenses](#)

No license ▾

Create repository

clone = save file locally

The screenshot shows the GitHub interface for a repository named 'TEST' by user 'wanyitang'. The repository is public and has 1 branch and 0 tags. The 'Code' button is highlighted with a green box, and its dropdown menu is open, showing options for cloning the repository. The 'HTTPS' option is highlighted with a red box, and the corresponding URL 'https://github.com/wanyitang/TEST.git' is shown. A green box also highlights the copy icon next to the URL. The repository description is 'This a test repository for LA's BeST program'. The right sidebar shows the 'About' section with a description, 'Readme', 'Activity', '0 stars', '0 watching', and '0 forks'. Below this are sections for 'Releases', 'Packages', and 'Contributors'.

wanyitang / TEST

Type / to search

<> Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

TEST Public

Pin Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file

Add file

<> Code

Local Codespaces

Clone

HTTPS SSH GitHub CLI

https://github.com/wanyitang/TEST.git

Clone using the web URL.

Open with GitHub Desktop

Download ZIP

About

This a test repository for LA's BeST program

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

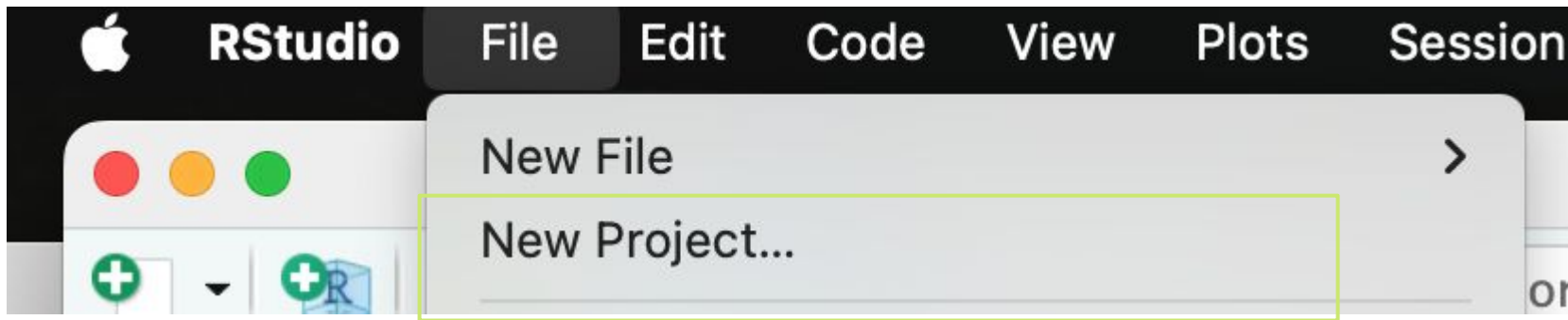
[Publish your first package](#)

Contributors 1

wanyitang

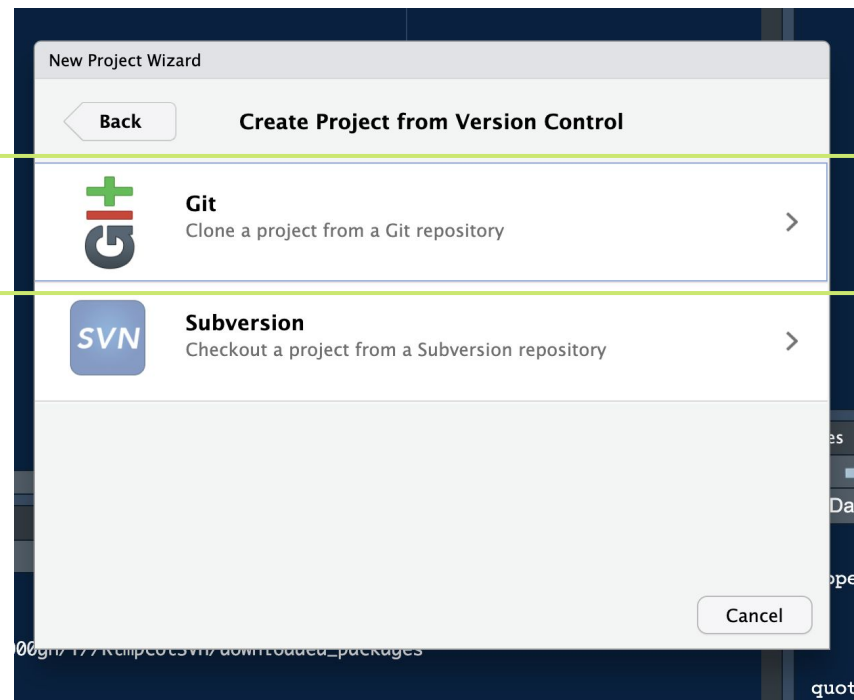
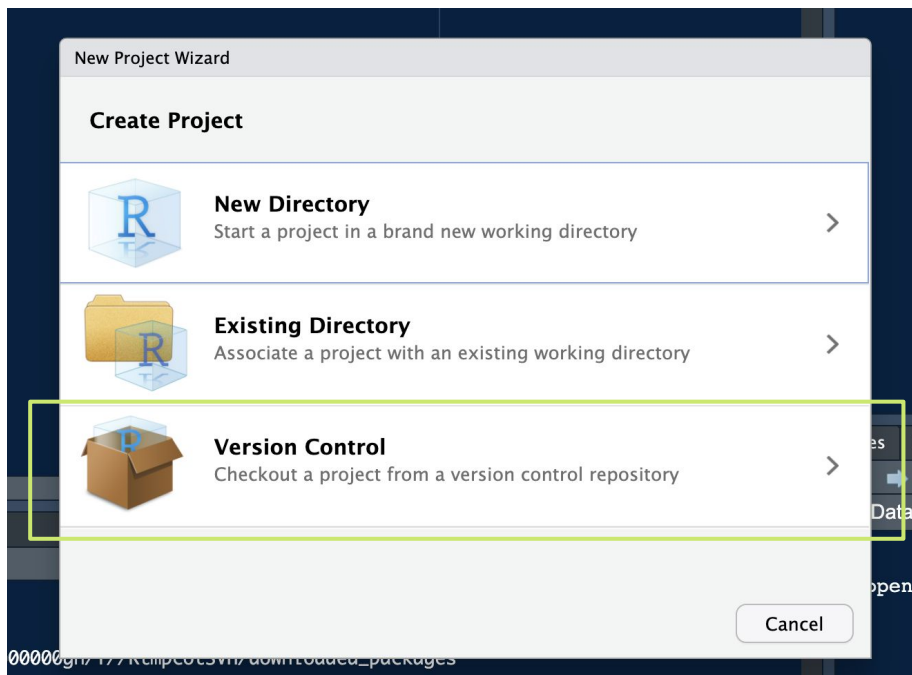
Use git with R studio

Step 1



A **project** is a folder that contains all the files for one piece of work, such as code, data, notes, and results.

Use Git with R studio



clone with R studio

New Project Wizard

Back **Clone Git Repository**



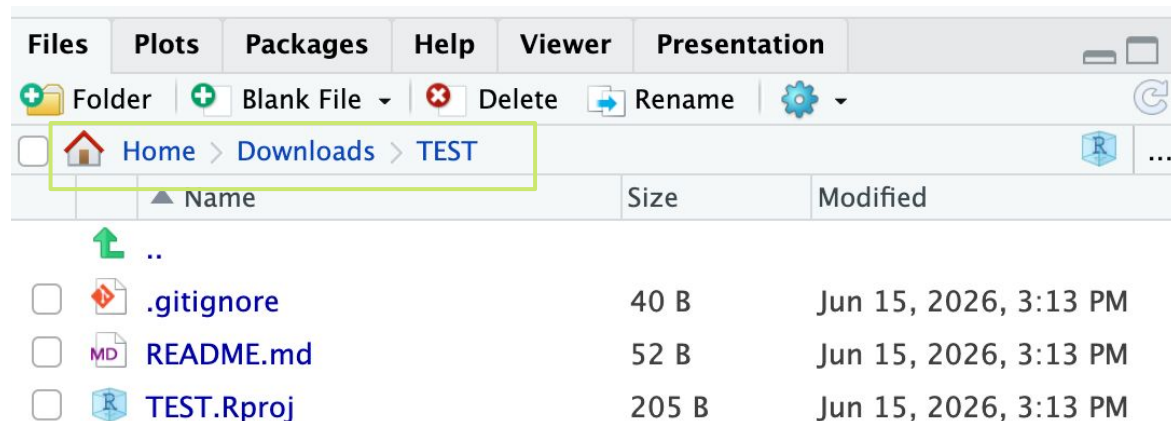
Repository URL:

Project directory name:
 Folder name

Create project as subdirectory of:
 Path

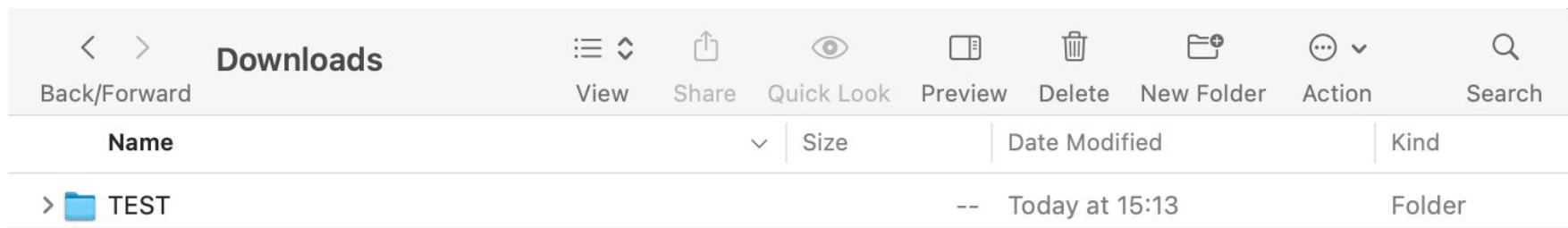
☐ Open in new session

Check your R files and local folder



The screenshot shows the RStudio interface with the 'Files' pane on the left. The breadcrumb path 'Home > Downloads > TEST' is highlighted with a green box. Below the path, a table lists the files in the 'TEST' folder:

	Name	Size	Modified
	..		
☐	.gitignore	40 B	Jun 15, 2026, 3:13 PM
☐	README.md	52 B	Jun 15, 2026, 3:13 PM
☐	TEST.Rproj	205 B	Jun 15, 2026, 3:13 PM

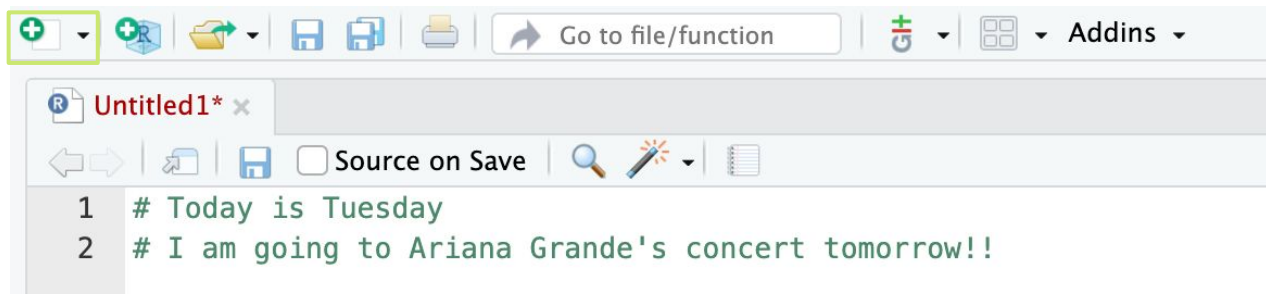


The screenshot shows a file explorer window titled 'Downloads'. The breadcrumb path is '> TEST'. The table below lists the contents of the 'TEST' folder:

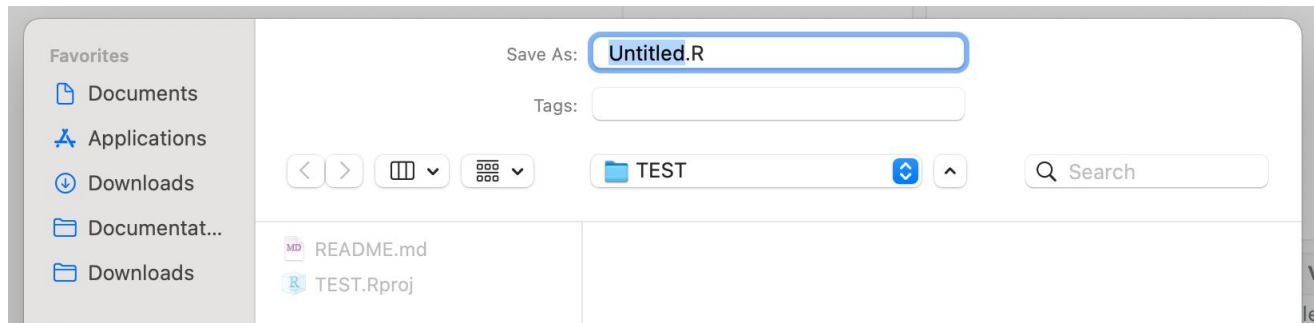
Name	Size	Date Modified	Kind
> TEST	--	Today at 15:13	Folder

commit = save checkpoint

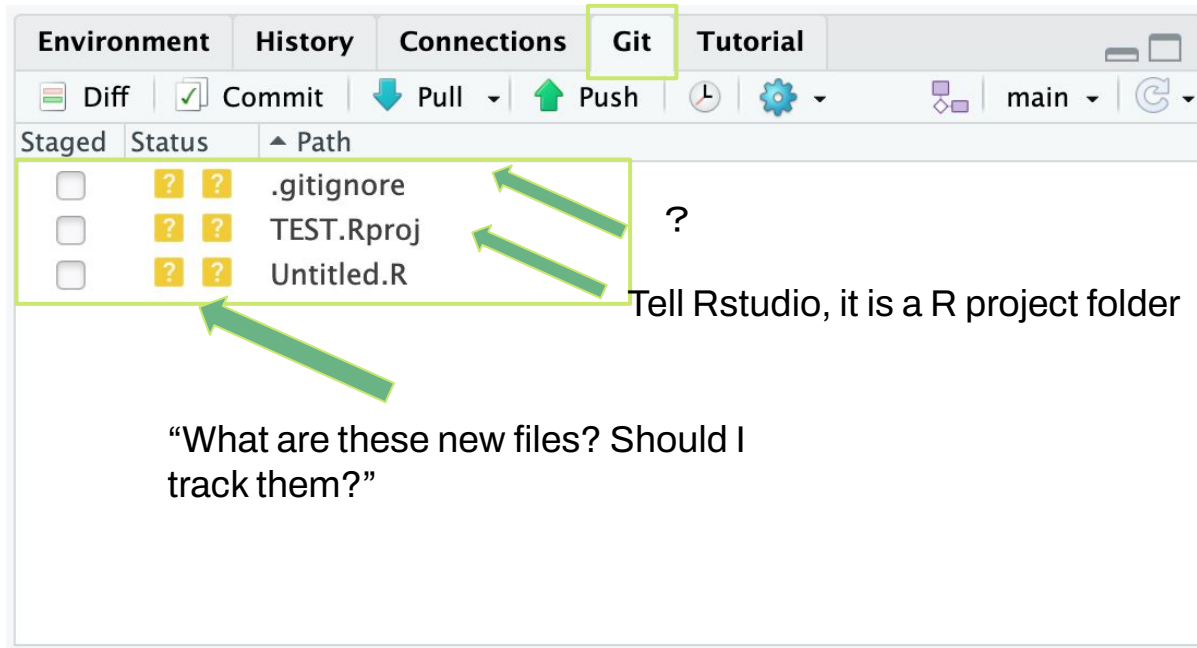
1. Create a new file



2. Save it. The path to your local git repository just pops up!



Find **git** and do your first **commit**!



The screenshot shows the RStudio Git interface. The top bar has tabs for Environment, History, Connections, Git (highlighted with a green box), and Tutorial. Below the tabs is a toolbar with buttons for Diff, Commit, Pull, and Push, along with icons for a clock, a gear, and a refresh button. The main area displays a table of files with columns for Staged, Status, and Path. The files listed are .gitignore, TEST.Rproj, and Untitled.R, each with a checkbox in the Staged column and a yellow question mark icon in the Status column. A green box highlights the first three rows of the table. Three green arrows point from the text 'Tell Rstudio, it is a R project folder' to the TEST.Rproj and Untitled.R rows. Another green arrow points from the text 'What are these new files? Should I track them?' to the .gitignore row.

Staged	Status	Path
☐	?	.gitignore
☐	?	TEST.Rproj
☐	?	Untitled.R

Tell Rstudio, it is a R project folder

“What are these new files? Should I track them?”

Do your first **commit**

Always write a nice message! save time. Save life. Save everything

The screenshot shows the RStudio 'Review Changes' interface. At the top, a toolbar contains buttons for 'Diff', 'Commit', 'Pull', 'Push', and settings. The 'Commit' button is highlighted with a green box. Below the toolbar, a table lists files with their status and paths:

Staged	Status	Path
☒	A	.gitignore
☒	A	TEST.Rproj
☒	A	Untitled.R

Green arrows point from the text 'Added' to the 'Staged' column and from 'Ready to commit' to the 'Status' column. Below this table, another table shows the same files, with the 'Untitled.R' file selected. To the right, the 'Commit message' field is active, containing the text 'Initial commit!!!'. A green arrow points from the text 'Always write a nice message!' to this field. At the bottom right, the 'Commit' button is highlighted with a green box. Below the commit message field, there are checkboxes for 'Amend previous commit' and 'Sign commit'. At the very bottom, a code editor shows the content of the selected file, 'Untitled.R':

```
@@ -0,0 +1,2 @@
1 # Today is Tuesday
2 # I am going to Ariana Grande's concert tomorrow!!
   No newline at end of file
```

Added

Ready to commit

Unstage chunk

Have we already upload the changes to Github?

Check your Github repository page!

Have we already upload the changes to Github?

 **TEST** Public

 Pin  Watch **0**  Fork **0**  Star **0**

 main  1 Branch  0 Tags T Add file <> Code

 **wanyitang** Initial commit a6c108f · 1 hour ago 🕒 1 Commit

 README.md Initial commit 1 hour ago

 **README**



TEST

This a test repository for LA's BeST program

About

This a test repository for LA's BeST program

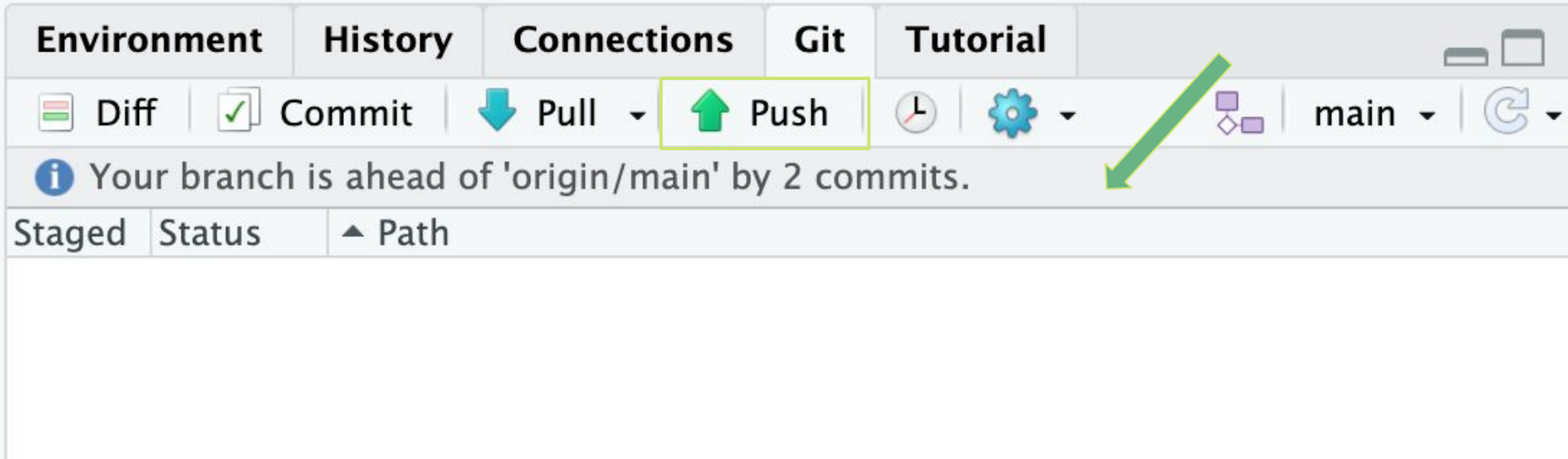
-  Readme
-  Activity
-  0 stars
-  0 watching
-  0 forks

Releases

No releases published

[Create a new release](#)

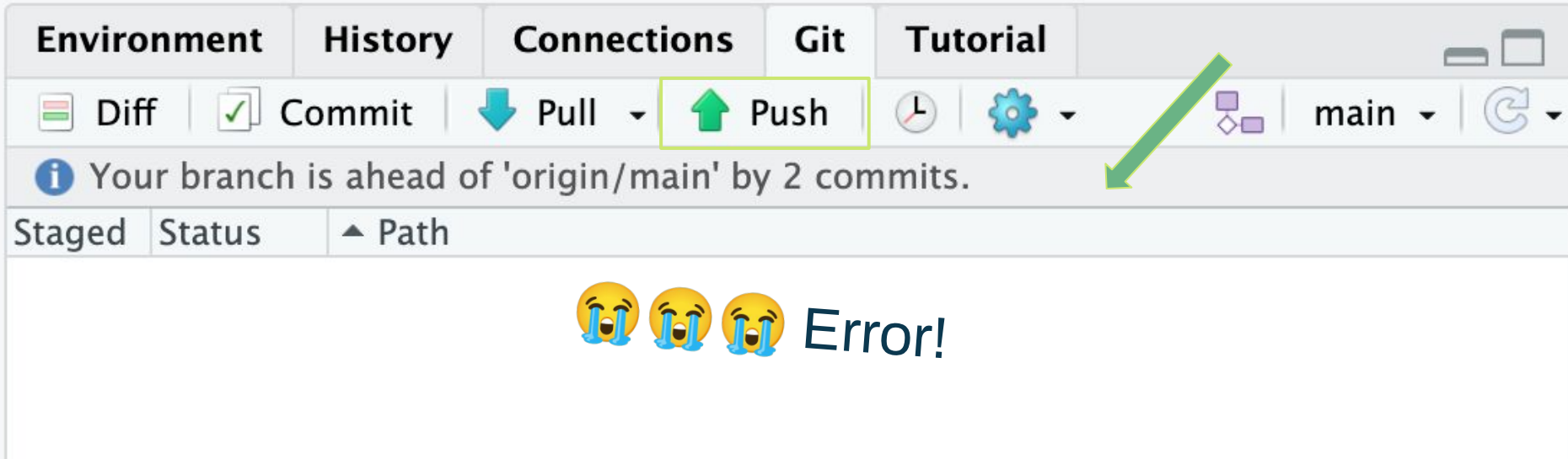
push = upload



The screenshot shows the top toolbar of a code editor with the following tabs: Environment, History, Connections, Git, and Tutorial. The 'Git' tab is active, displaying a row of icons: Diff, Commit, Pull, Push, a clock, a gear, a branch icon, and a refresh icon. The 'Push' button, represented by a green upward arrow, is highlighted with a green rectangular box. A green arrow points from the right side of the interface towards the 'Push' button. Below the toolbar, a status bar displays the message: "Your branch is ahead of 'origin/main' by 2 commits." The bottom section of the interface has a header with "Staged", "Status", and "Path" columns, followed by a large empty table area.

Environment	History	Connections	Git	Tutorial
Diff	Commit	Pull	Push	
Your branch is ahead of 'origin/main' by 2 commits.				
Staged	Status	Path		

push = upload



Now GitHub does not accept account passwords for Git push.
For HTTPS, we need to use a personal access token

In R Console:

```
install.packages("usethis")
```

```
install.packages("gitcreds")
```

```
usethis::create_github_token()
```

Settings / Developer Settings

GitHub Apps

OAuth Apps

Personal access tokens

Fine-grained tokens

Tokens (classic)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

FOR R studio

☐ **admin:ssh_signing_key** Full control of public user SSH signing keys

☐ **write:ssh_signing_key** Write public user SSH signing keys

☐ **read:ssh_signing_key** Read public user SSH signing keys

Generate token Cancel

In R Console:

gitcreds::gitcreds_set()

Then copy from github and paste your token in R

Personal access tokens (classic)

Generate new token ▾

Tokens you have generated that can be used to access the [GitHub API](#).



Make sure to copy your personal access token now. You won't be able to see it again!

✓ ghp_jr00VvK6TnpSdpaTQIf8p0YpanFsCI13LSED 

Delete

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

```
> gitcreds::gitcreds_set()
```

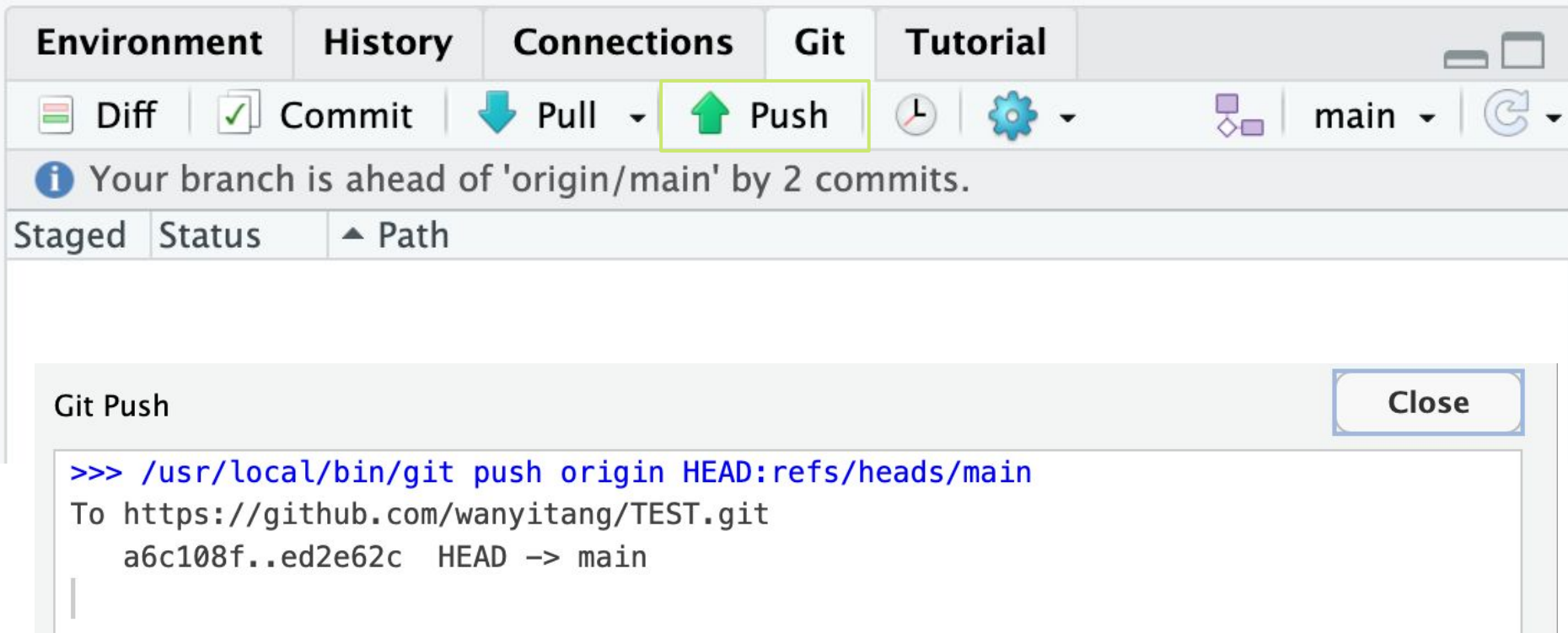
```
? Enter password or token: ghp_jr00VvK6TnpSdpaTQIf8p0YpanFsCI13LSED
```

```
-> Adding new credentials...
```

```
-> Removing credentials from cache...
```

```
-> Done.
```

push = upload

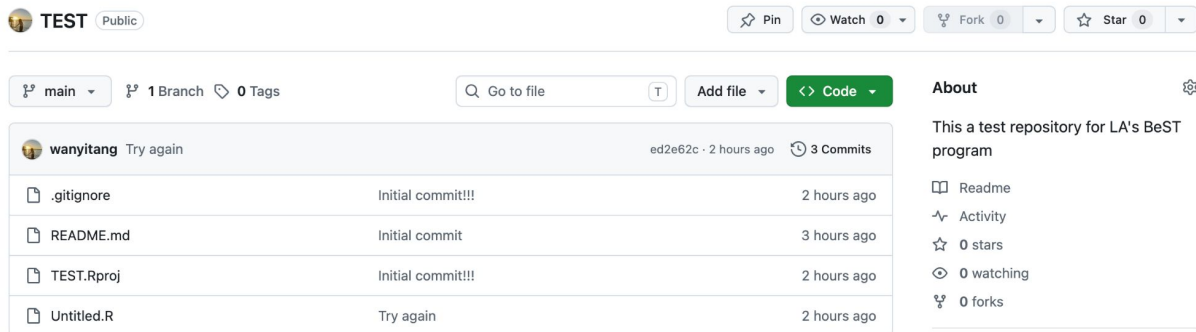


The screenshot shows a Git GUI interface with a top bar containing tabs: Environment, History, Connections, Git, and Tutorial. Below the tabs is a row of buttons: Diff, Commit, Pull, Push, a clock icon, a gear icon, a branch icon, and a refresh icon. The 'Push' button is highlighted with a green border. Below the buttons is a status bar that reads: "Your branch is ahead of 'origin/main' by 2 commits." Below the status bar is a table with columns: Staged, Status, and Path. Below the table is a terminal window titled "Git Push" with a "Close" button. The terminal displays the following text:

```
>>> /usr/local/bin/git push origin HEAD:refs/heads/main
To https://github.com/wanyitang/TEST.git
   a6c108f..ed2e62c  HEAD -> main
```

Have we already upload the changes to Github?

Check your Github repository page again!



The screenshot shows a GitHub repository named 'TEST' (Public) by user 'wanyitang'. The repository has 1 branch (main) and 0 tags. The commit history shows four commits: '.gitignore' (Initial commit!!!, 2 hours ago), 'README.md' (Initial commit, 3 hours ago), 'TEST.Rproj' (Initial commit!!!, 2 hours ago), and 'Untitled.R' (Try again, 2 hours ago). The repository description is 'This a test repository for LA's BeST program'. The repository has 0 stars, 0 watching, and 0 forks.

File	Commit Message	Time
.gitignore	Initial commit!!!	2 hours ago
README.md	Initial commit	3 hours ago
TEST.Rproj	Initial commit!!!	2 hours ago
Untitled.R	Try again	2 hours ago



pull = download

Let's make some change on github page first

The screenshot shows the GitHub web editor interface. At the top, the file path is 'TEST / README.md' in the 'main' branch. On the right, there are two buttons: 'Cancel changes' and 'Commit changes...'. The 'Commit changes...' button is highlighted with a green box. Below the editor, a modal dialog titled 'Commit changes' is open. It has a 'Commit message' field with the text 'Update README with additional weather comment' and a refresh icon. Below that is an 'Extended description' field with the placeholder text 'Add an optional extended description...'. At the bottom of the dialog, there are two radio button options: 'Commit directly to the main branch' (which is selected) and 'Create a new branch for this commit and start a pull request'. A link 'Learn more about pull requests' is provided below the second option. At the bottom of the dialog, there are two buttons: 'Cancel' and 'Commit changes'. The 'Commit changes' button is highlighted with a green box.

TEST / README.md in main

Cancel changes Commit changes...

Edit Preview

Spaces 2 Soft wrap

```
1 # TEST
2 This a test repository for LA's BeST program
3 I love today's weather!
4
```

Commit changes

Commit message

Update README with additional weather comment

Extended description

Add an optional extended description...

Message and description suggested by Copilot.

☒ Commit directly to the main branch

☐ Create a new branch for this commit and start a pull request

[Learn more about pull requests](#)

Cancel Commit changes

Do we need to push on github?

pull = download

Let's make some change on github page first

The screenshot shows the GitHub web editor interface. At the top, the file path is 'TEST / README.md' in the 'main' branch. On the right, there are two buttons: 'Cancel changes' and 'Commit changes...'. The 'Commit changes...' button is highlighted with a green box. Below the file path, there are tabs for 'Edit' and 'Preview'. The 'Edit' tab is active, showing the content of README.md:

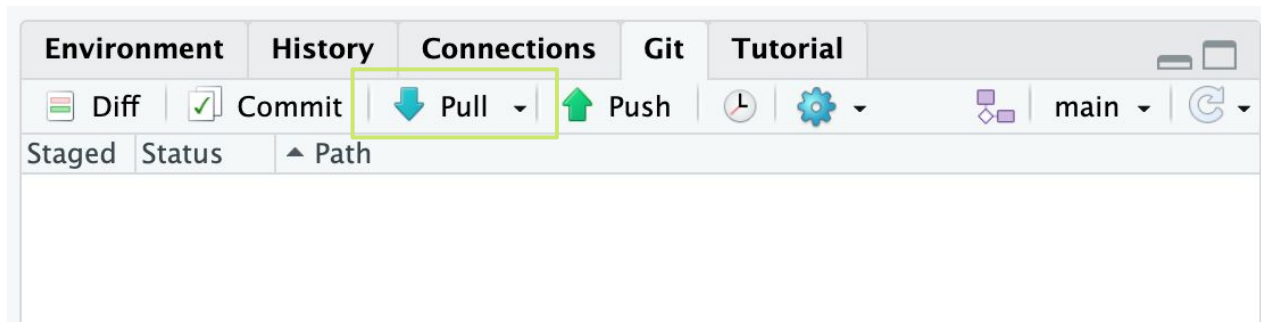
```
1 # TEST
2 This a test repository for LA's BeST program
3 I love today's weather!
4
```

A 'Commit changes' dialog box is open in the center. It has a title bar with a close button. Inside, there is a 'Commit message' section with a text input field containing 'Update README with additional weather comment' and a refresh icon. Below that is an 'Extended description' section with a text area containing 'Add an optional extended description...'. At the bottom of the dialog, there is a note 'Message and description suggested by Copilot.' and two radio button options: 'Commit directly to the main branch' (which is selected) and 'Create a new branch for this commit and start a pull request'. A link 'Learn more about pull requests' is provided for the second option. At the bottom of the dialog, there are two buttons: 'Cancel' and 'Commit changes'. The 'Commit changes' button is highlighted with a green box.

Do we need to push on github?

A: Imagine github as a google drive, you don't need to upload as you are already here

pull = download



pull = download

Check your changes!



The screenshot shows a code editor window with two tabs: 'Untitled.R' and 'README.md'. The 'README.md' tab is active. The toolbar includes navigation arrows, a 'Preview on Save' checkbox, a green checkmark icon, a magnifying glass icon, a 'Preview' button with a dropdown arrow, and a settings gear icon. Below the toolbar, there are two tabs: 'Source' (selected) and 'Visual'. The 'Source' tab displays the following text:

```
1 # TEST
2 This a test repository for LA's BeST program
3 I love today's weather!
4
```

Use clone to save others' file

<https://github.com/USCbiostats/LAsBEST>

New Project Wizard

Back Clone Git Repository



Repository URL:

Project directory name:
 Folder name

Create project as subdirectory of:
 Path

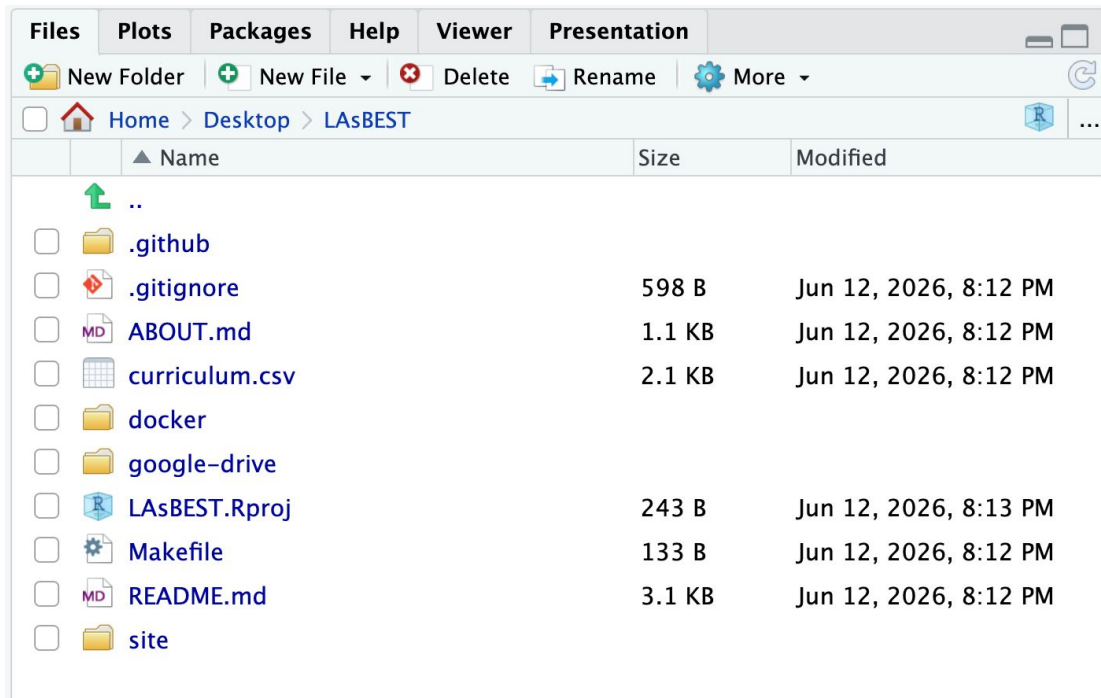
☐ Open in new session

- Clone the USCbiostats/LAsBEST repository from GitHub.
- Create a local project folder at **~/YOUR_PATH/LAsBEST**.
- Open the project in RStudio.
- Verify that the project name LAsBEST appears in the upper-right corner of the RStudio window.

clone with R studio

<https://github.com/USCbiostats/LAsBEST>

Check the R files in R studio!



- Clone the USCbiostats/LAsBEST repository from GitHub.
- Create a local project folder at **~/YOUR_PATH/LAsBEST**.
- Open the project in RStudio.
- Verify that the project name LAsBEST appears in the upper-right corner of the RStudio window.

clone with R studio

Check the folder!

<https://github.com/USCbiostats/LAsBEST>

LASBEST			
Name	Date Modified	Size	Kind
 ABOUT.md	Today at 8:12 PM	1 KB	Markdown File
 curriculum.csv	Today at 8:12 PM	2 KB	Comm...et (.csv)
>  docker	Today at 8:12 PM	--	Folder
>  google-drive	Today at 8:12 PM	--	Folder
 LASBEST.Rproj	Today at 8:13 PM	243 bytes	R Project
 Makefile	Today at 8:12 PM	133 bytes	Plain Text
 README.md	Today at 8:12 PM	3 KB	Markdown File
>  site	Today at 8:12 PM	--	Folder

***Try collaboration
using **FORK** if have
time**

What we got using fork:

📖 README



class-2026-introduction

Introduce yourself!

Im marcelo perez guzman from lawrence, massachusetts

im T Smith a junior mathematics major with statistical concentration and mass communications minor

My name is Briza Tayagua-Delgado!

i love cheesecake

My name is Alan!

Liliana!

hi i'm Jorden

hello, I'm sydney

This is ameya

This is Cadence!

🍴 14 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 12



Other resources

- Book Happy Git and Github for the useR:
 - <https://happygitwithr.com/big-picture>
- Video tutorial to set-up Git and Github in Rstudio:
 - <https://rfortherestofus.com/2021/02/how-to-use-git-github-with-r>
- Youtube video explaining Git basics:
 - <https://youtu.be/eWxxfttcMts>
- Paper: Excuse me, do you have a moment to talk about version control?
 - <https://peerj.com/preprints/3159v2/#>